



Building a Sample for Time Series Analysis

For the Inference of Macro Models

Gauthier Vermandel

Summer School on Macroeconomic Modelling and Policy

Objectives

- ▶ Downloading data in real time in MATLAB;
- ▶ Getting a stationary sample for macroeconomic analysis.

Additional reading list

- ▶ Nelson and Plosser(1982)
- ▶ Canova(2007)
- ▶ Smets and Wouters(2007)
- ▶ Uhlig(1995)

Course materials (slides, code, problem set):

github.com/Vermandel/Macro-Modelling-Course-Nancy

Back in time

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- ▶ Influential revolution, including in macroeconomic theory: pushed toward much parsimonious models (Lucas' Revolution).
- ▶ Development of a rich set of time series models: AR, MA, VAR, ARCH, ... up to DSGE models (that we will estimate later on).

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- ▶ As new data are released, applied economists must re-run their forecasting models taking into account the latest observation released.
- ▶ In recent times, this policy work operates with real time data to facilitate the repetitive work of the econometrician.
- ▶ We will use DB-nomics to get data in real time.

Reading list:

I assume you have a good knowledge in standard time series models, the reading list can be shortened to:

- ① Basic assessment of the stationarity in MATLAB: ADF Test
- ② A guide for DBnomics.

Outline

Introduction

Data crafting

- Building a ready-to-use sample

- Data in real time

- Estimation Routine

Statistical Evaluation of DSGE

- Numerical solution of a DSGE model

- Inference of a DSGE model

Core assumption in conventional models

- ▶ A time series Y_t is a process in sequence over time $t \in \{1, 2, \dots, T\}$:

$$\{y_1, y_2, \dots, y_T\}$$

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- ▶ How was this sample generated?
- ▶ Understanding how this sample was generated is a fundamental task for an applied economist.
- ▶ By identifying the **underlying data-generating process**, we can develop meaningful quantitative analyses (e.g. causality analysis, forecasting, counterfactuals, etc).

Core assumption in conventional models

- One can assume a specific **Data Generating Process** (i.e. a time series model) with specific distribution for prediction errors. These models typically relate the present value of a series to past values (AR model) and past prediction errors (MA):

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- ▶ ... but under certain conditions (stationarity, gaussian errors, linearity) the computation of $E(y_{t+h}|I_t)$ is **straightforward**.
- ▶ In this class, our focus will be only on **linear gaussian time series models**, which imply to be careful on the features of the sample we utilize.

Core assumption in conventional models

But which kind of features should be look for?

- **Stationarity** (weak definition) typically allows the mean function $E(y_t)$ and auto-covariance function $E[(y_s - E(y_s))(y_t - E(y_t))]$ to be time independent.

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- ▶ One can estimate the parameters of the time series model $\hat{f}(\cdot)$, and provide out-of-sample forecast assuming distribution of y_t **remains unchanged** over the forecasting horizon (very convenient).

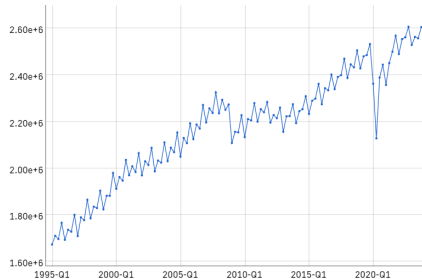
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- ▶ Suppose our time series model is $y_t = \mu + \phi y_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim N(0, \sigma^2)$. One can estimate $\hat{\mu}$, $\hat{\phi}$ and $\hat{\sigma}$, and compute the mean-wise forecast $E(y_{t+1}|I_t) = \hat{\mu} + \hat{\phi}Y_t$ and quantify uncertainty: $E(y_{t+1}|I_t) \pm 1.96 \cdot \hat{\sigma}$.

But in practice...

Most of time series are affected by trend and seasonal components that do **not** imply (weak) stationarity. See GDP time series:



Gross domestic product at market prices – Euro area

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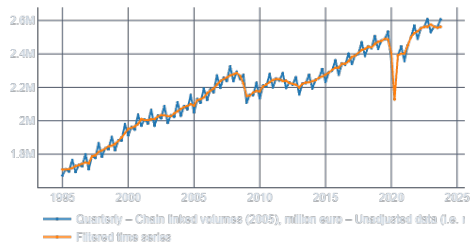
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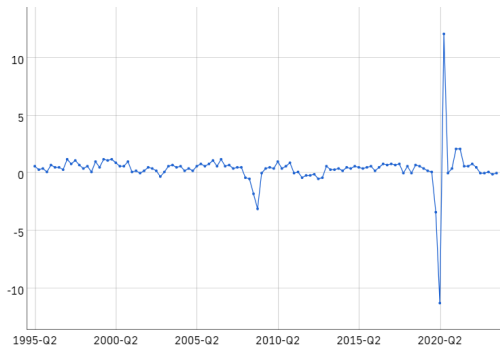
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- ② **Check for stationarity** (through visual screening and stationarity tests), and if necessary, impose stationarity:
 - ▶ Use first difference filtering (e.g., express in growth rates)
 - ▶ Use a business cycle filter (HP, bandpass, etc)
 - ▶ Eventually divide by working age population to get rid off low frequency component

First deseasonalize:



Deseasonalized gross domestic product at market prices – Euro area

Then impose stationarity $\Delta \log Y_t$



Deseasonalized GDP growth at market prices – Euro area

DBnomics: Real-Time Data for Forecasting

What is dbnomics?

dbnomics is a platform that aggregates economic datasets from various public sources.

Why is it useful for forecasting?

- ▶ Provides access to a vast array of real-time economic data.
- ▶ Allows economists and analysts to track economic indicators as they are released.
- ▶ Enables more accurate and timely forecasting models by incorporating the latest data.

DBnomics: Real-Time Data for Forecasting

Scenario: INSEE French GDP Data Release

- ▶ INSEE releases French GDP data every first Monday of the month at 10:00 AM.
- ▶ As a forecaster, you only need to rerun your code at 10:01 AM to access the latest data.
- ▶ This allows you to quickly generate graphs and draft reports for management.

Using dbnomics

- ▶ Access the latest French GDP data from INSEE via dbnomics API.
- ▶ Integrate the new data into your forecasting model.
- ▶ Generate updated graphs and reports for management with minimal delay.

DBnomics: Real-Time Data for Forecasting

How to use dbnomics?

- 1 Go to db.nomics.world
- 2 Browse among all the available databases, find your time series, and pick its token (a unique identifier for each series):

United States - Gross domestic product - expenditure approach - Volume index, OECD reference year, seasonally adjusted - Annual

[OECD/QNA/USA.B1_GE.VIXOBSA.A]



min: 11.621 max: 119.02 avg: 53.023 σ : 32.126

Add to cart Download 

DBnomics: Real-Time Data for Forecasting

How to use dbnomics?

3 Download DBnomics MATLAB plugging `call_dbnomics.m`

4 Query dbnomics on real time (assuming you have internet connection):

```
1 [output_mat,output_table,dates_nb] = call_dbnomics(varargin);
```

Input:

`varargin`: string tokens list ('token1','token2',...)

Outputs:

`output_mat`: $T \times (N + 1)$ matrix of extracted data, 1st column is date vector in MATLAB format.

`output_table`: same matrix but in table format.

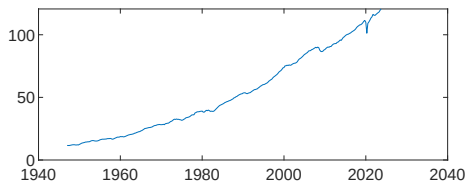
`dates_nb`: vector of dates in numeric values.

DBnomics: Real-Time Data for Forecasting

How to use dbnomics?

Example:

```
1 [output_mat,output_table,dates_nb] = call_dbnomics('OECD/QNA/  
   USA.B1_GE.VIXOBSA.Q');  
2 figure;  
3 plot(dates_nb,output_mat(:,2))
```



DBnomics: Real-Time Data for Forecasting

Building my sample for the US

Suppose a New Keynesian model comprising 3 core variables $[y_t, \pi_t, r_t]$. Our gross data from dbnomics comprise a price data $\mathbb{P}_t$, GDP (volume) $\mathbb{Y}_t$ and nominal rate $\mathbb{R}_t$. One can simply impose for the two stationary variables the following transformation:

$$G_{\mathbb{Y}t} = \log(\mathbb{Y}_t) - \log(\mathbb{Y}_{t-1})$$

$$G_{\mathbb{P}t} = \log(\mathbb{P}_t) - \log(\mathbb{P}_{t-1})$$

One can next map the model and the data as follows:

$$y_t - y_{t-1} = G_{\mathbb{Y}t}$$

$$\pi_t = G_{\mathbb{P}t}$$

$$\underbrace{r_t}_{\text{model}} = \underbrace{\mathbb{R}_t}_{\text{data}}$$

DBnomics: Real-Time Data for Forecasting

Let's have a look at the code...

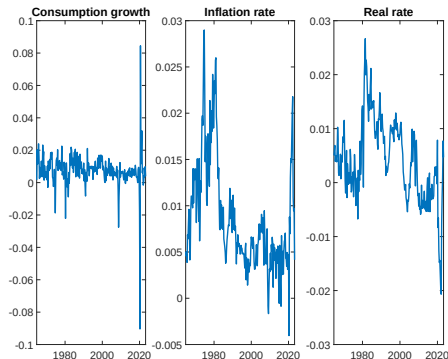
```
1 [raw_table,~,dates_nb] = call_dbnomics(...  
2     'OECD/QNA/USA.B1_GE.CQRSA.Q',...  
3     'OECD/QNA/USA.B1_GE.DNBSA.Q',...  
4     'OECD/KEI/IR3TIB01.USA.ST.Q');  
5  
6 valid_idx = all(~isnan(raw_table(:,2:end)),2);  
7 P = raw_table(valid_idx,3);  
8 P = P / P(find(floor(dates_nb(valid_idx)) == 2015,1));  
9 gy_obs = diff(log(raw_table(valid_idx,2) ./ P));  
10 pi_obs = diff(log(P));  
11 r_obs = raw_table(valid_idx,4) / 400;
```

Volume GDP identifies real activity, the deflator gives inflation, and the nominal rate is used directly in the sample.

DBnomics: Real-Time Data for Forecasting

Building my sample for the US

Run `my_db_US.m` to get the model observables:



A quick recap for your report:

- ▶ We will employ linearized and stationary macro models, which implies to have a sample distribution consistent with our model.
- ▶ Check first that your data do not exhibit seasonal component → use deseasonalized series or applies methods to remove seasonal component (e.g. X13 method).
- ▶ Check your sample is stationary (visual screening to assess the data have no upward trend as for GDP). If so, employ first order difference.
- ▶ If stationary and no seasonal features, your sample is ready for applied work.

A general formulation

- ▶ Consider a structural model written in a state-space form:

$$\mathbb{E}_t\{f_{\Theta}(y_{t-1}, y_t, y_{t+1}, \eta_t)\} = 0$$

where:

- ▶ y_t vector $N_y \times 1$ of endogenous variables;
- ▶ η_t vector $N_{\eta} \times 1$ of gaussian innovations, $\eta_t \sim \mathcal{N}(0, \Sigma)$.
- ▶ $f_{\Theta}()$ set of nonlinear equations.
- ▶ Θ vector of structural parameters.
- ▶ $\mathbb{E}_t\{\cdot\}$ the expectation scheme.

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- ▶ Spoilert alert: we will use 3rd one!

A general formulation

- ▶ Consider a function $f(x)$. A Taylor expansion around a fixed point a reads as:

$$f(x) \simeq f(a) + \frac{f'(a)}{1!} (x - a) + \frac{f''(a)}{2!} (x - a)^2 + \dots$$

- ▶ Two natural questions:
 - 1 Which order of approximation to consider?
 - 2 Which value of a ?

A first order taylor expansion

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- ▶ F , G , H and M are matrices stacking first order derivatives of f_Θ
- ▶ $\hat{z}_t = y_t - a$ is typically expressed in distance from steady state $a=y$. Why? linear model combined with gaussian stochastic shocks $\eta_t \sim \mathcal{N}(0, \Sigma)$ implies that $y_t - y \sim \mathcal{N}(0, \Sigma_y)$. The smallest deviations $E((y_t - a)^2)$ are obtained when 1st order expansion computed around steady state.

A first order taylor expansion

- ▶ One looks for a recursive solution that would locally:

$$\hat{z}_t = P\hat{z}_{t-1} + Q\eta_t$$

where P and Q are two unknown matrices.

- ▶ We refer to `slide_2.pdf` for a description of the solution.

History of modern macroeconometrics

- ▶ Back in the 2 000s, dominance of time series models (e.g. VAR) to make forecasts and historical decomposition.
- ▶ DSGE models were not yet fully exploited on empirical grounds to compete with a-theoretical models.
- ▶ Estimation of DSGE models were based on weak information method: matching moments.
- ▶ Few attempts to use full-information techniques: Fair and Taylor(1983) & Ireland(2004), but subject to identification issues.

Principles of likelihood techniques

- ▶ Consider a specific Data Generating Process described by our DSGE model:

$$\hat{z}_t = P(\theta) \hat{z}_{t-1} + Q(\theta) \eta_t \text{ with } \eta_t \sim \mathcal{N}(0, \Sigma)$$

$$Y_t = H(\theta) \hat{z}_t + v_t, \text{ with } v_t \sim \mathcal{N}(0, R)$$

- ▶ Where $\theta \subset \Theta$ is the subset of parameters to be estimated, the remaining being calibrated.
- ▶ Subset of observable variables $Y_t \subset \hat{z}_t$, recognized as the set of macroeconomic time series, and map the model's equations into where v_t are measurement errors.

Principles of likelihood techniques

- ▶ Consider a sample $Y_1, Y_2, \dots, Y_T$, one can compute the a prediction error:

$$S_t(\theta) = Y_t - E_{t-1}\{Y_t(\theta)\}$$

where the prediction errors variance $\Omega_t = E(S_t S_t')$.

- ▶ Unlike VAR, in DSGE there are more endogenous variables than observable ones
→ use Kalman filter to update recursively by picking 'best' prediction in the Kalman filter through optimal mean squared error (MSE) estimator.
- ▶ The likelihood function is given by:

$$\log p(Y_{1:T}|\theta) = -\frac{nT}{2} \log(2\pi) - \frac{1}{2} \sum_1^T \log(|\Omega_t|) - \frac{1}{2} \sum_1^T S_t' \Omega_t^{-1} S_t \quad (1)$$

Principles of likelihood techniques

- ▶ So in principle, (i) compute prediction errors (i.e. sequence of shocks) $S_{\theta,t}$ that replicates a sample $Y_{1:T}$, (ii) calculate likelihood function, (iii) relative comparison of $\log p(Y_{1:T}|\theta')$ against $\log p(Y_{1:T}|\theta'')$ allow to determine which θ'/θ'' is the most likely to have generated the sample.
- ▶ Optimization can be implemented to determine θ_{MLE} that is the value of θ in the population of parameters that is the most likely to have generated the sample:

$$\min_{\{\theta\}} -\log p(Y_{1:T}|\theta)$$

- ▶ As long as $\log p(Y_{1:T}|\theta)$ is differentiable in θ , one can use gradient based methods to optimize.

Limits of likelihood techniques

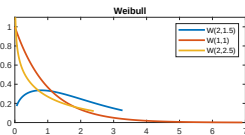
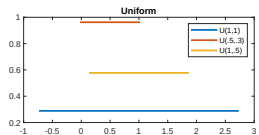
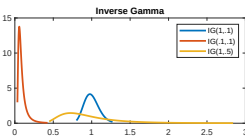
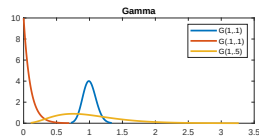
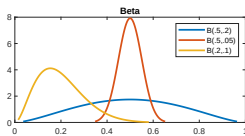
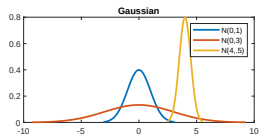
- ▶ Any model is an approximate of the true data generating process.
- ▶ Does it matter? If the data-generating distribution does not belong to the model's set of probability distributions, then the model is misspecified.
- ▶ **Some illustrations:** our current medium-scale NK model does not include a financial sector or epidemic dynamics → recessions will be captured erroneously by wrong sources of fluctuations.
- ▶ Likelihood function will be higher for implausible parameter values to accomodate the wrong representation of the data given by our misspecified model.

Setting Priors

- Dynare includes a set of priors, each prior is summarized by a shape (e.g. gaussian, beta, etc.) and a mean and standard deviation.
- How to adjust my prior? Usually have a look at already published papers, or use common sense.

Shape	Name	Support	Example(s)
normal_pdf	$\mathcal{N}(\mu, \sigma)$	$(-\infty, +\infty)$	Policy parameters, utility curvatures.
gamma_pdf	$\mathcal{G}(\mu, \sigma)$	$(0, +\infty)$	Elasticities, utility curvatures, trends.
beta_pdf	$\mathcal{B}(\mu, \sigma)$	$[0, 1]$	Probabilities, AR-MA terms, shares.
inv_gamma_pdf	$\mathcal{IG}_1(\mu, \sigma)$	$(0, +\infty)$	Shocks standard deviations.
uniform_pdf	$\mathcal{U}(\mu, \sigma)$	$(-\infty, +\infty)$	Set lower/upper bounds.
weibull_pdf	$\mathcal{W}(\mu, \sigma)$	$[0, +\infty)$	Measurement error std.
inv_gamma2_pdf	$\mathcal{IG}_2(\mu, \sigma)$	$(0, +\infty)$	Standard deviations.

Setting Priors



Estimation steps:

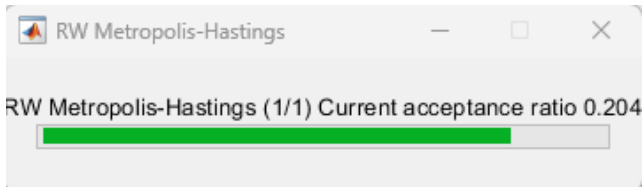
- 1 Maximize $p(\theta|Y_{1:T})$ using numerical optimization techniques to find the mode of the distribution.
- 2 Use numerical integration to compute confidence intervals (parametric uncertainty) through Metropolis Hastings algorithm.

Each iteration requires: for a given candidate θ to compute prediction errors $\{S_{\theta,t}\}_1^T$, evaluate sample likelihood + posterior.

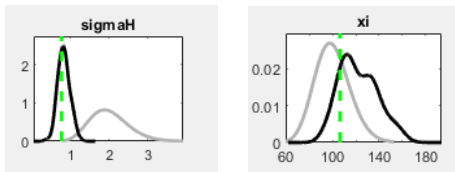
Dynare automatizes those steps, making life easier!

Acceptation

- ▶ Metropolis-Hastings looks to explore the parameter space. Sampling: $\theta_i^{cand} \sim \mathcal{N}(\theta_{i-1}^{cand}, c \times \Sigma_\theta)$ with parameter c that scale how large random perturbation around the mode.
- ▶ c must be set to match 20-30% of acceptance rate.
- ▶ If acceptance rate too low: chain stuck into local minimum. If too low: chain does not explore parameter space sufficiently.



Parameter Identification



- ▶ Dark line: posterior distribution $p(Y_{1:T}|\theta)p(\theta)$, while gray line is $p(\theta)$ for alternative parameter values θ . Green line is mean.
- ▶ Left figure: data are informative (data dominate); right figure: data are uninformative (prior dominates).
- ▶ Identification implies that prior does not dominate posterior. Here, we should calibrate ξ and re-run estimation.

Toy DSGE: model and data

Economic structure

We use the simplest possible two-equation log-linearised system:

$$\underbrace{gc_obs_t}_{\text{cons. growth}} = \sigma \underbrace{rr_obs_t}_{\text{real rate}}, \quad rr_obs_t = \rho_r rr_obs_{t-1} + \varepsilon_t^r.$$

σ is the slope of the IS curve; ρ_r governs persistence of the real rate.

- ▶ Observable 1: **consumption growth** (gc_obs), built from real consumption in `my_db_US.m`.
- ▶ Observable 2: the **real rate** (rr_obs), proxied by $r_t^r \approx r_t - \pi_{t+1}$.
- ▶ Estimated parameters: ρ_r , σ , σ_{ε^r} , and a measurement-error std on gc_obs .

Toy DSGE: priors and estimation

Estimated parameters

In `simple_euler_real_rate.mod`, four parameters are estimated: ρ_r , σ , σ_{ε^r} , and the measurement-error std on `gc_obs`.

- ▶ ρ_r : **Beta**($\mu = 0.80$, $\sigma = 0.10$) — persistence in $[0, 1]$, real rate is usually persistent.
- ▶ σ : **Gamma**($\mu = 2$, $\sigma = 0.20$) — IS slope, strictly positive, tight prior around 2.
- ▶ σ_{ε^r} : **inv-Gamma**($\mu = 0.01$, $s = 2$) — structural shock std, strictly positive.
- ▶ σ_{gc_obs} : **Weibull**($\mu = 0.001$, $\sigma = 0.001$) — measurement-error std on consumption growth.

Pedagogical objective: connect data treatment, observables, priors, and likelihood-based estimation in the smallest possible Dynare example.

A simple Dynare exercise

```
1 var gc_obs rr_obs;
2 varexo eps_r;
3 parameters rho_r sigma;
4
5 rho_r = 0.8; sigma = 1.0;
6
7 model(linear);
8     gc_obs = sigma * rr_obs;
9     rr_obs = rho_r * rr_obs(-1)
10         + eps_r;
11 end;
12 varobs gc_obs rr_obs;
```

```
1 estimated_params;
2     rho_r,      beta_pdf ,
3         0.8,    0.10;
4     sigma,      gamma_pdf ,
5         2,      0.20;
6     stderr eps_r, inv_gamma_pdf
7         , 0.01,  2;
8     stderr gc_obs, weibull_pdf ,
9         0.001, 0.001;
10 end;
11
12 estimation(datafile=myobs ,
13     prefilter=1, mode_compute=4,
14     mh_replic=0, diffuse_filter)
15 ;
```

Takeaways

Data

- ▶ Check for **stationarity** (visual screening + ADF test); apply first-difference or HP filter if needed.
- ▶ Check for **seasonality**; use a seasonal filter (e.g. X-13ARIMA-SEATS) if present.
- ▶ Use **real-time data** (e.g. DBnomics) to update the dataset in real time.

Estimation

- ▶ Set **priors** consistent with the economic literature and calibration targets.
- ▶ Monitor the **acceptance rate** of the Metropolis-Hastings sampler (target: 20–30%).
- ▶ Check **parameter identification**: posterior should dominate the prior.



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Bayesian techniques

Bayes' Theorem

$$\underbrace{p(\theta|Y)}_{\text{posterior}} = \frac{\overbrace{p(Y|\theta)}^{\text{likelihood}} \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(Y)}_{\text{marginal density}}}$$

- ▶ $p(Y|\theta)$: probability of observing the data given parameters — the **likelihood function**, evaluated via the Kalman filter.
- ▶ $p(\theta)$: **prior** distribution on θ , encoding beliefs before seeing the data.
- ▶ $p(\theta|Y)$: **posterior** distribution — what we learn about θ after observing Y .
- ▶ $p(Y) = \int p(Y|\theta) p(\theta) d\theta$: **marginal data density**, a normalising constant

Out-of-Sample Forecasts

- ▶ Governments and central banks rely on DSGE model forecasts to design monetary and fiscal policies
- ▶ A forecast provides a baseline scenario, and allow to introduce some policies to evaluate how the forecast is affected

A quick recap. Suppose that we have picked $\hat{\theta}$ with highest posterior

$$\hat{z}_t = P(\hat{\theta}) \hat{z}_{t-1} + Q(\hat{\theta}) \eta_t$$

with corresponding estimated sequence of shocks $\{\eta_t\}_{t=1}^T$ for given $\hat{\theta}$, one wants to make some out of sample forecast over horizon $T + h$

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